

PILOT: Guiding Long-Horizon World-Model Planning with Visual Subgoals and Policy Priors for Robotic Manipulation

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Abstract—World-model planning for long-horizon manipulation is constrained by short reliable prediction horizons: a distant goal may provide little guidance for preparatory actions, while uninformed action sampling makes local optimization inefficient. We propose PILOT (Policy-Initialized Local Optimization with Temporal subgoals), which uses a video model to generate intermediate visual targets and a learned policy to initialize the Cross-Entropy Method (CEM). An action-conditioned world model refines candidate actions toward each target in a receding-horizon control loop. Our primary evaluation fine-tunes Wan and π_0 on the same 40 real-robot demonstrations. Across four UR5e manipulation tasks, PILOT achieves 68.8% success, compared with 55.0% without policy initialization, 13.8% for policy-only execution, and 0.0% for direct final-goal planning. These results show that visual subgoals learned from a small demonstration collection can support effective online control even when direct policy execution remains unreliable, with policy initialization further improving local search efficiency. Complementary simulation experiments show gains of 30 and 22 percentage points over direct execution of the same policy on long Push-T tasks and on LIBERO-10.

I. INTRODUCTION

World models[1], [2], [3], [4] offer a flexible paradigm for robotic control by enabling actions to be optimized at test time through predictive simulation. Rather than committing to a fixed policy, a robot can use an action-conditioned dynamics model to evaluate candidate behaviors and adapt its decisions online. This capability is particularly appealing for goal-conditioned manipulation, where the desired outcome may vary across tasks and environments. However, despite recent progress in visual world models and latent-space planning, extending such methods to long-horizon manipulation remains challenging. Many real-world tasks require a sequence of temporally ordered interactions—for example, approaching an object, grasping it, transporting it, and placing it at a target—whose intermediate stages are not explicitly specified by the final goal.

Most world-model planners formulate manipulation as a goal-conditioned optimization problem: candidate action sequences are rolled out through the learned dynamics model and selected according to their predicted distance to a desired goal observation. While effective over short horizons, this formulation becomes increasingly difficult as the task horizon grows. A distant goal image specifies *what* the final scene should look like, but provides little information about *how the task should progress* toward that state. The planner must therefore infer both the intermediate interaction sequence and the low-level actions needed to realize it. In a non-convex planning landscape, this ambiguity can lead

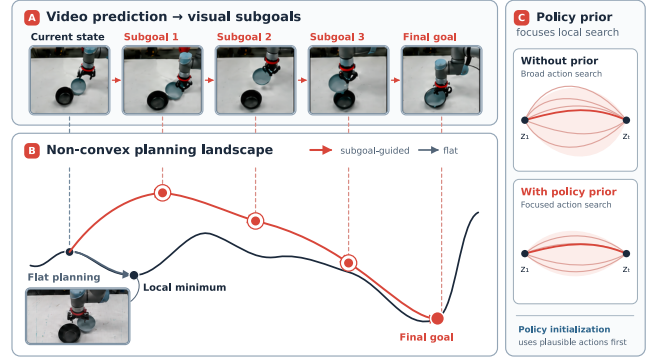


Fig. 1. Motivation of PILOT. Visual subgoals decompose long-horizon planning over a non-convex objective into easier local planning stages, while a learned policy prior focuses action search on plausible trajectories to improve local planning efficiency.

to shortcut solutions or locally favorable trajectories that reduce visual goal distance without completing the intended manipulation. Longer model rollouts further amplify this difficulty through accumulated prediction error.

The task horizon and the prediction horizon play different roles in this problem. A long task can be executed through many short MPC updates, while model error and computation constrain how far each update can reliably predict. Within such a short rollout, an action needed to prepare for a later interaction may not reduce distance to the final goal. The objective can therefore favor short-sighted behavior even when the local action search is effective. Visual subgoals address this issue by exposing intermediate progress within the available prediction horizon.

A second challenge arises from action search. Sampling-based optimizers such as the Cross-Entropy Method (CEM) avoid differentiating through the dynamics model, but their efficiency depends on the region of action space being explored. Uninformed sampling spends computation on implausible or irrelevant trajectories, and the search space grows rapidly with prediction horizon. World-model planning therefore needs guidance both over *where the task should progress* and over *where the optimizer should search*.

Our key insight is that these two difficulties can be addressed by introducing structure along complementary axes. Temporal structure can be provided by intermediate visual targets that decompose a distant goal into a sequence of shorter planning problems. Recent video prediction models offer a natural mechanism for generating such targets by forecasting temporally coherent future observations. At the same time, behavioral structure can be introduced through

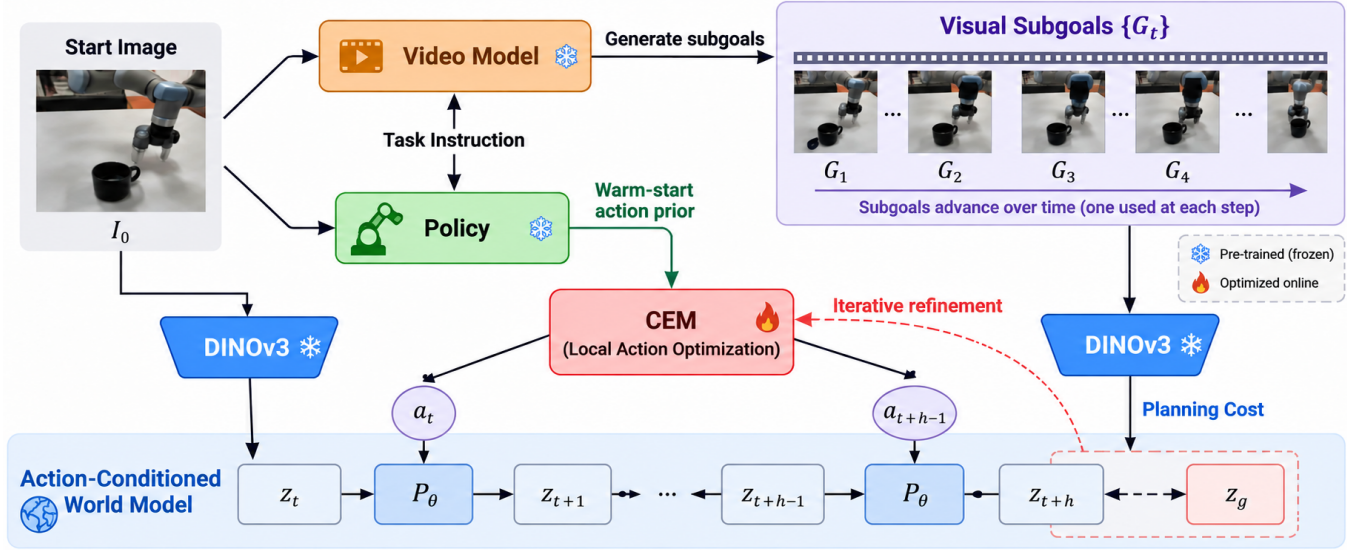


Fig. 2. Overview of PILOT. A video prediction model generates a sequence of visual subgoals from the current observation and task instruction, providing temporal guidance for long-horizon task progression. For each active subgoal, a learned policy prior initializes CEM around behaviorally plausible action sequences, while an action-conditioned world model evaluates and refines candidate actions online in latent space.

a learned policy prior that concentrates sampling around plausible action trajectories. Importantly, neither component needs to replace model-based planning: the video model only proposes intermediate task progress, while the policy only provides an initialization for action search. Feasibility and action selection remain the responsibility of the action-conditioned world model during online optimization.

Based on this perspective, we propose **PILOT** (Policy-Initialized Local Optimization with Temporal subgoals), a hierarchical framework for guiding non-convex world-model planning in long-horizon robotic manipulation. PILOT separates long-horizon task progression from short-horizon action optimization. A video world model first generates a sequence of temporally consistent visual subgoals that expose intermediate stages of the task and decompose the overall objective into local planning problems. For each active subgoal, a learned policy prior initializes CEM around behaviorally plausible action sequences, reducing the amount of uninformed search. An action-conditioned world model then evaluates and refines these candidate actions online toward the active visual target.

Our primary experimental setting is adaptation from a small collection of real-robot demonstrations. We fine-tune Wan and π_0 on the same 40 expert demonstrations and ask whether their learned knowledge can be used effectively through different roles: visual task progression for the generator and action proposals for the policy. A video prediction can be useful as a local planning target even when a policy trained on the corresponding actions is not reliable enough to execute the entire task directly. The action-conditioned world model supplies the online action evaluation needed to turn those targets into control.

We evaluate PILOT on four real-world long-horizon manipulation tasks using a UR5e robot platform. PILOT achieves a 68.8% average task success rate, outperform-

ing direct final-goal world-model planning and policy-only execution. Our ablations show that visual subgoal decomposition is critical for overcoming the difficulty of direct long-horizon optimization, while policy-prior initialization further improves average task success and makes local CEM optimization more efficient.

Simulation complements the real-world evaluation with controlled analyses over more initial states. Push-T varies task duration at a fixed prediction horizon; both Push-T and LIBERO-10 include final-goal planning with policy initialization and an expert-subgoal reference. These comparisons test whether action proposals compensate for missing temporal guidance, how prior strength trades off against planning budget, and how subgoal source affects execution. Wan receives additional video fine-tuning data for adaptation to rendered scenes, so the shared small-demonstration comparison is specific to the real-world setting.

The contributions of this work are summarized as follows:

- We identify two complementary sources of difficulty in long-horizon world-model planning: insufficient temporal guidance over intermediate task progress and inefficient search over a large, non-convex action space.
- We propose **PILOT**, which combines video-predicted subgoals, policy-initialized CEM, and online action refinement to execute long tasks through short-horizon world-model planning.
- We evaluate PILOT on real-world robotic manipulation tasks and analyze the distinct roles of visual subgoals and policy-prior initialization in improving task success and online planning efficiency, and show in simulation that even weak priors improve the success-versus-planning-time trade-off.

A. Related Work

a) World-Model Planning for Robotic Manipulation:

Recent work has demonstrated that visual world models can support robotic planning by predicting future observations or latent states under candidate actions. DINO-WM [5] learns action-conditioned dynamics over frozen DINOv2[6] patch features and performs test-time optimization by matching predicted latent states to a visual goal. V-JEPA 2 [7] similarly leverages large-scale self-supervised video pretraining and trains an action-conditioned predictor on robot interaction data for image-goal manipulation. LeWorldModel [8] explores a complementary direction by learning a compact JEPA-style latent world model end-to-end from pixels, with the goal of improving planning efficiency without relying on a frozen foundation encoder.

These methods establish latent visual prediction as a basis for goal-conditioned control. PILOT adds a temporal hierarchy to this formulation: a video model proposes intermediate targets, and the action-conditioned world model solves local planning problems toward them.

b) *Visual Subgoals for Long-Horizon Planning:* Long-horizon manipulation[9], [10], [11] is difficult to specify using only a final goal observation because the intermediate stages of task execution are often underdetermined. One approach is to provide additional intermediate observations. V-JEPA 2 [7], for example, can plan toward image subgoals, but such subgoals must be available externally and therefore require additional task-specific specification or data collection. Language provides a more convenient task-level interface, but does not directly encode the fine-grained visual changes required during manipulation.

PILOT instead uses a video prediction model to translate the current visual scene and language instruction into a sequence of concrete intermediate visual targets. The generated trajectory is not treated as an executable motion plan; rather, it provides temporal guidance for downstream model-based control. This distinction matters because visual plausibility and action feasibility need not be resolved by the same model. Compared with HWM [12], which generates intermediate latent goals within a hierarchical world model, PILOT uses interpretable image-space subgoals as high-level guidance while leaving action evaluation and control to a separate action-conditioned world model online.

c) *Policy Priors for Model-Based Planning:* Sampling-based optimizers such as CEM [13] and MPPI [14] are attractive for world-model planning because they do not require differentiating through the learned dynamics. Their performance, however, depends strongly on the distribution from which candidate action sequences are sampled. In long-horizon manipulation, uninformed initialization can waste substantial computation on trajectories that are behaviorally implausible or unrelated to the task.

Recent work has therefore explored learned proposal distributions for model-based planning. WorldPlanner [15] uses a diffusion-based action sampler to generate structured candidate actions for planning, while PiJEPA [16] warm-starts model-based optimization using a learned policy

prior in visual navigation. PILOT adopts the same general principle of informed initialization, but integrates it with visual-subgoal-conditioned manipulation planning. The policy is not executed as the controller; instead, it provides a task-conditioned proposal around which CEM performs local optimization. The action-conditioned world model then evaluates and refines these actions toward the active visual subgoal.

II. METHOD

We consider goal-conditioned robotic manipulation from visual observations. At each decision step, an action-conditioned world model is used within a model-predictive control (MPC) loop to evaluate candidate action sequences, execute a short prefix of the optimized sequence, observe the resulting state, and replan.

PILOT introduces structure into this optimization process along two complementary axes. At the task level, a video prediction model provides *temporal guidance* by generating intermediate visual subgoals that decompose a long-horizon objective into a sequence of local planning problems. At the action level, a learned policy prior provides *search guidance* by initializing CEM around behaviorally plausible action sequences. The action-conditioned world model then remains responsible for online evaluation and refinement toward the active subgoal. An overview of the framework is shown in Fig. 2.

A. Problem Formulation

At time t , the robot observes an image o_t , which is encoded into a latent state

$$s_t = \phi(o_t). \quad (1)$$

An action-conditioned world model predicts the latent dynamics

$$\hat{s}_{t+1} = f_\theta(s_t, a_t), \quad (2)$$

where $a_t \in \mathcal{A}$ denotes the robot action.

Given a visual goal g , we measure progress using the latent-space distance

$$c(s, g) = \|s - \phi(g)\|_2^2. \quad (3)$$

In our implementation, the encoder ϕ and action-conditioned dynamics model f_θ are instantiated with JEPA-WM [17].

Let T_{task} denote the task duration in control steps, and h the prediction horizon of a single planning call. These quantities are distinct: long-horizon manipulation can require $h \ll T_{\text{task}}$, with many replanning calls needed to complete the task. Model error and computation constrain the prediction horizons that are useful in practice.

A conventional visual MPC planner directly optimizes an h -step action sequence toward the final goal:

$$a_{t:t+h-1}^* \approx \arg \min_{a_{t:t+h-1} \in \mathcal{A}^h} c(\hat{s}_{t+h}, g), \quad (4)$$

where $\hat{s}_{t+h}$ is obtained by recursively applying Eq. (2) along the candidate action sequence.

For long-horizon manipulation, this formulation requires the optimizer to simultaneously discover intermediate task progress and low-level control through a single final-goal objective. It also couples longer model rollouts with search over a high-dimensional action sequence. PILOT instead decomposes the task into a sequence of subgoal-conditioned local optimization problems. Each local problem is solved over a short prediction horizon $h \ll T_{\text{task}}$, while a learned policy prior initializes the action search before world-model refinement.

Under a limited prediction horizon, a necessary preparatory action may not reduce final-goal distance within the rollout. This can make the objective favor short-sighted actions even if the action sequence is optimized accurately. Sampling limitations introduce a separate difficulty: CEM may fail to find useful actions for the current target. Our controlled comparisons use the same h for flat and subgoal-guided world-model planning, so the benefit of changing targets is evaluated without increasing model rollout length.

B. Temporal Visual Subgoals

PILOT uses a video prediction model as a high-level temporal proposal mechanism. Importantly, the video model is neither action-conditioned nor used to evaluate candidate controls. Its role is to predict a coarse visual progression of the task and expose intermediate states that can serve as local planning targets.

We instantiate the video module with Wan2.1-1.3B[18] and fine-tune it on task videos collected in the robot environment. Given the current observation o_t and a language instruction ℓ , the model predicts a future visual trajectory containing T_v frames:

$$V(o_t, \ell) = \{\tilde{o}_1, \tilde{o}_2, \dots, \tilde{o}_{T_v}\}. \quad (5)$$

From the predicted trajectory, we select K visual subgoals

$$\begin{aligned} \mathcal{G} &= \{g_1, g_2, \dots, g_K\}, \\ g_k &= \tilde{o}_{\tau_k}, \end{aligned} \quad (6)$$

where $\tau_k \in \{1, \dots, T_v\}$ denotes the selected frame index. Unless otherwise stated, the indices are uniformly spaced along the predicted rollout.

These generated frames are used only as intermediate visual targets. PILOT does not require the video rollout itself to correspond to an executable robot trajectory. Instead, the video model provides a coarse temporal decomposition of the task, while action feasibility is assessed online by the action-conditioned world model during local optimization. This separation allows the high-level model to propose *where the task should progress* without requiring it to predict the precise control sequence needed to reach each state. Generated targets can nevertheless be unreachable or poorly timed; the expert-subgoal comparisons in Sec. III-F evaluate the effect of replacing these targets under the same controller.

C. Policy Prior for Local Action Search

Temporal subgoals make the target more local while preserving the short prediction horizon h , but each local problem

still requires optimization over robot actions. Standard CEM initialization does not exploit task-specific behavioral structure and may therefore allocate many samples to implausible action sequences. PILOT addresses this issue using a learned policy prior π_ψ .

In our implementation, π_ψ is obtained by fine-tuning a π_0 policy on expert demonstrations. The policy is conditioned on the current observation o_t and task instruction ℓ :

$$a_{t:t+h-1}^\pi \sim \pi_\psi(\cdot \mid o_t, \ell), \quad (7)$$

where the first h predicted actions form the local action proposal.

The policy is used only as a *prior for optimization*; its predicted actions are not executed directly. Instead, $a_{t:t+h-1}^\pi$ initializes the local CEM search in a behaviorally plausible region of the action space. The policy is also not conditioned on the generated visual subgoal g_k . Consequently, explicit alignment with the active subgoal is supplied by the world-model objective, while the policy can still encode task-level behavioral information. This design separates behavioral initialization from goal-directed online refinement: the policy suggests *where to search*, while the action-conditioned world model determines which candidate actions best progress toward the current visual target. The Flat WM + prior comparison tests whether this behavioral initialization can compensate for the absence of intermediate visual targets.

D. Policy-Initialized Local Optimization

PILOT combines the video prediction model V , policy prior π_ψ , and action-conditioned world model f_θ in a hierarchical receding-horizon planning procedure, summarized in Algorithm 1. At the beginning of the planning episode, the video model generates a sequence of visual subgoals. The controller then processes these subgoals sequentially. For each active subgoal, PILOT repeatedly solves a short-horizon local optimization problem and executes only a prefix of the optimized action sequence before replanning.

For subgoal g_k , the local planner optimizes an h -step action sequence according to

$$J_k(a_{t:t+h-1}; s_t) = c(\hat{s}_{t+h}, g_k), \quad (8)$$

where $s_t = \phi(o_t)$ is the current latent state and $\hat{s}_{t+h}$ is obtained by rolling out Eq. (2) for h steps.

The learned policy proposal initializes the CEM distribution as

$$\mu_0 = a_{t:t+h-1}^\pi, \quad \Sigma_0 = \Sigma_{\text{init}}. \quad (9)$$

CEM then samples candidate action sequences from the current distribution, evaluates them using the action-conditioned world model and the subgoal cost in Eq. (8), selects low-cost candidates, and refits the sampling distribution. This iterative refinement allows the planner to deviate from the policy proposal whenever the world model predicts that alternative actions better reach the active subgoal.

After optimization, the mean of the final CEM distribution is taken as the plan a^* , and only its first $h_{\text{exec}} < h$ actions are executed. The robot then receives a new observation and

replans, preserving the closed-loop nature of MPC. In the real-world experiments, we use a local planning horizon of $h = 2$ and execute $h_{\text{exec}} = 1$ action at each control step.

For each visual subgoal, we allocate a fixed budget of R planning–execution cycles before advancing to the next subgoal. We use $R = 2$ in the real-world experiments. This fixed-budget design avoids introducing an additional subgoal-completion model or manually tuned latent-distance threshold, while keeping the high-level progression mechanism identical across tasks.

A complete traversal of K targets executes KRh_{exec} control steps unless terminated early. The fixed schedule does not adapt target timing when execution falls behind. The simulation evaluation is described in Sec. III-F.

Algorithm 1: PILOT: Policy-Initialized Local Optimization with Temporal Subgoals

Input: current observation o_t , task instruction ℓ , video model V , policy prior π_ψ , world model f_θ

Hyperparameters: $K, h, h_{\text{exec}}, \Sigma_{\text{init}}, R$, episode step limit

- 1) $\tilde{O} \leftarrow V(o_t, \ell)$
 - 2) $\mathcal{G} = \{g_1, \dots, g_K\} \leftarrow \text{SELECTSUBGOALS}(\tilde{O})$
 - 3) For $k = 1, \dots, K$:
 - a) For $r = 1, \dots, R$:
 - i) $s_t \leftarrow \phi(o_t)$
 - ii) $a_{t:t+h-1}^\pi \sim \pi_\psi(\cdot | o_t, \ell)$
 - iii) $(\mu_0, \Sigma_0) \leftarrow (a_{t:t+h-1}^\pi, \Sigma_{\text{init}})$
 - iv) $a^* \leftarrow \text{CEM}(J_k, f_\theta, s_t, \mu_0, \Sigma_0)$
 - v) Execute $a_{t:t+h_{\text{exec}}-1}^*$ and observe $o_{t+h_{\text{exec}}}$
 - vi) $t \leftarrow t + h_{\text{exec}}$
 - vii) Stop if the task is complete or the episode step limit is reached.
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III. EXPERIMENTS

We first evaluate task success and planning efficiency on a real UR5e, followed by qualitative executions. Push-T and LIBERO-10 provide complementary analyses of task length, policy initialization, prior strength versus planning budget, and subgoal source.

A. Real-World Setup

We use a UR5e with a fixed RGB camera (Fig. 3) on four tabletop tasks: *Cup*, *Pencil Case*, *Bowl-Box*, and *Bowl-Bowl*. Each involves approaching, grasping or contacting, transporting, and placing an object. We evaluate each method for 20 trials per task and report per-task success counts and average success rate.

Training data and implementation. We fine-tune Wan2.1-1.3B [18] and π_0 [19] on the same 40 successful demonstrations, 10 per task, using videos for Wan and observation–action trajectories for π_0 . JEPA-WM is trained on these demonstrations together with approximately five hours of random-play interaction collected roughly evenly across the four settings. Thus, the matched demonstration budget concerns task-specific fine-tuning of Wan and π_0 ; system training also includes world-model interaction data and foundation-model pretraining. A trial succeeds when the robot completes the specified manipulation within the execution horizon.

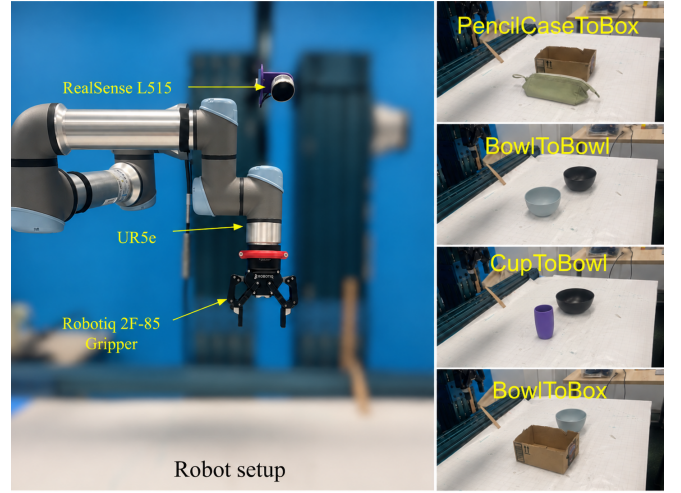


Fig. 3. Real-world UR5e evaluation setup. We evaluate PILOT on four long-horizon tabletop manipulation tasks using visual observations from a fixed RGB camera.

TABLE I

REAL-WORLD TASK SUCCESS RATES OVER 20 TRIALS PER TASK. AVG. POOLS 80 TRIALS WITH 95% WILSON CONFIDENCE INTERVALS.

Method	Pencil				Avg.(95% CI)
	Cup	Case	Bowl-Box	Bowl-Bowl	
Flat WM	0/20	0/20	0/20	0/20	0.0% (0.0–4.6)
Policy only	4/20	3/20	2/20	2/20	13.8% (7.9–23.0)
PILOT w/o prior	8/20	5/20	14/20	17/20	55.0% (44.1–65.4)
PILOT	11/20	13/20	13/20	18/20	68.8% (57.9–77.8)

All world-model variants share the dynamics model, action bounds, latent cost, episode allowance, and CEM budget, with $h = 2$ and $h_{\text{exec}} = 1$. Generated-subgoal variants use $K = 6$ targets and $R = 2$ planning–execution cycles per target. Without the policy prior, CEM starts from a zero-mean Gaussian over the same action bounds.

B. Baselines and Ablations

We evaluate four core methods on the real robot and in simulation:

- **Flat WM:** CEM plans directly toward the final goal image using the learned action-conditioned world model, without intermediate visual subgoals or policy-prior initialization. It uses the same short prediction horizon h as PILOT.
- **Policy only:** The learned task-conditioned policy is executed directly, without online world-model evaluation or CEM refinement.
- **PILOT w/o prior:** The planner uses the same generated visual subgoals as PILOT, but initializes CEM from a generic distribution rather than from the learned policy prior.
- **PILOT:** The full method combines video-predicted visual subgoals, policy-prior initialization, and online action refinement with the action-conditioned world model.

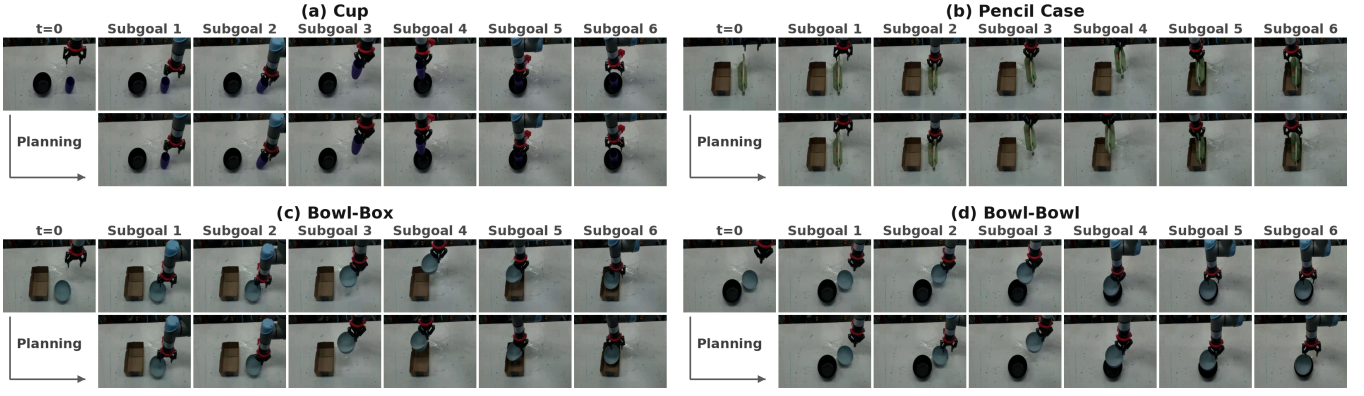


Fig. 4. PILOT on the four real-world tasks. Top row of each panel: initial observation and video-predicted visual subgoals. Bottom row: the robot’s observation when each subgoal expires.

C. Real-World Results

Table I shows that Flat WM completes none of the evaluated trials under the tested planning budget, while Policy only achieves 13.8% success. Adding visual subgoals to world-model planning raises success to 55.0% at the same prediction horizon. This improvement supports the value of intermediate objectives when the final goal provides insufficient guidance within a short rollout. The human-rated completion rate of the generated videos is 100% on all four tasks. With Wan and π_0 fine-tuned on the same demonstrations, the generated targets support useful online control even though direct policy execution remains unreliable.

PILOT reaches 68.8% success, a 13.8 percentage-point gain over PILOT w/o prior. The benefit varies by task: policy initialization adds three successes on Cup, eight on Pencil Case, and one on Bowl-Bowl, with one fewer on Bowl-Box. Its largest gain is therefore on Pencil Case. These comparisons establish the prior’s contribution when subgoals are available; Push-T additionally tests the prior with a final-goal objective.

D. Real-World Planning Efficiency

We vary the online planning budget and measure success against mean planning time. Policy initialization improves success under constrained budgets and reaches strong performance with less search (Fig. 5(a)). The corresponding CEM curves show a lower initial cost and lower cost throughout refinement (Fig. 5(b)), linking the execution gains to more effective local optimization.

E. Qualitative Analysis of Hierarchical Planning

Fig. 4 shows one execution per task: the top row of each panel is the video-predicted subgoals and the bottom row is the robot’s observation when each subgoal expires. The executed states track the subgoals while every planning call keeps the same short prediction horizon.

F. Simulation Setup

Environments. *Push-T* [20] is a planar pushing task in which a circular end-effector must push a T-shaped block onto a fixed target pose. It is a single-task, contact-rich

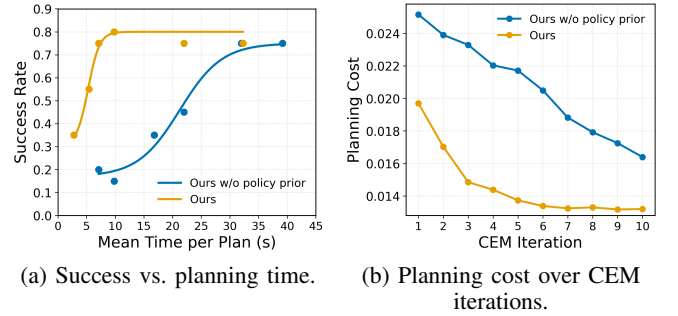


Fig. 5. Effect of policy-prior initialization on local planning efficiency. (a) Policy-guided initialization achieves higher task success under limited online planning budgets. (b) The prior provides a lower-cost initialization and maintains lower planning cost during CEM refinement.



Fig. 6. PILOT on Push-T. Top: initial observation and video-predicted visual subgoals. Bottom: the simulator observation when each subgoal expires.

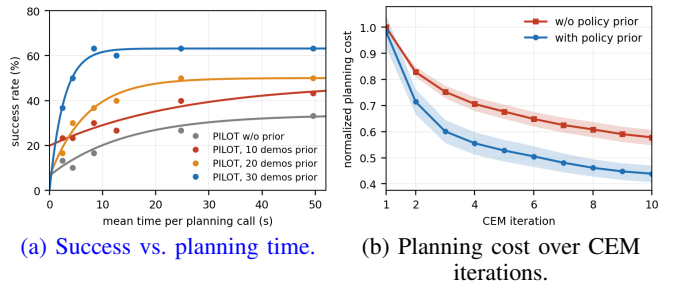


Fig. 7. Push-T planning efficiency. (a) Success on medium-length tasks versus mean time per planning call as the CEM budget varies, for generic initialization and for priors fine-tuned on 10, 20, and 30 demonstrations; lines are saturating-exponential fits. (b) CEM objective over iterations with and without policy initialization on identical planning problems from 90 held-out initial states; costs are normalized to generic initialization at iteration one, and shading shows standard errors.

environment with a two-dimensional action space, and allows the task horizon to be varied in a controlled way. *LIBERO-10* [21] contains ten language-conditioned, two-stage tabletop manipulation tasks on a Franka arm with a seven-dimensional operational-space action space, and is representative of realistic long-horizon manipulation.

Models and adaptation regime. Wan2.1-1.3B and π_0 receive environment-specific fine-tuning. Policy learning can be effective in simulation, particularly on LIBERO, while Wan requires additional video fine-tuning data to adapt to rendered scenes. Simulation therefore evaluates the planning framework under a different adaptation regime from the shared small-demonstration real-world setting. We fine-tune π_0 on expert action trajectories: on Push-T, π_0 uses 30 demonstrations (5 to 30 for the Policy only rows) and Wan 100; on LIBERO-10, π_0 uses 5 and Wan 20 demonstrations per task. For JEPA-WM, we use the released Push-T checkpoint and train a LIBERO model on benchmark demonstrations together with noise-perturbed replays.

Both simulation environments include two additional diagnostic variants:

- **Flat WM + prior:** The final-goal planner uses the same policy proposal and initial CEM covariance as PILOT, but retains the final goal throughout the episode. Together with Flat WM, PILOT w/o prior, and PILOT, this completes the 2×2 comparison of final-goal versus subgoal objectives and generic versus policy initialization.
- **PILOT (oracle):** The full planner replaces generated subgoals with frames from a held-out expert trajectory starting at the same evaluated initial state. The policy prior and world model are unchanged. This reference measures the effect of substituting expert temporal guidance; it is not assumed to be a strict upper bound on task success.

G. Push-T: Scaling with Task Horizon

We group Push-T initial states by expert episode length T_{expert} , measured in environment steps: *short* ($T_{\text{expert}} < 100$), *medium* ($100 \leq T_{\text{expert}} < 130$), and *long* ($T_{\text{expert}} \geq 130$). The observed ranges are 29–93, 110–129, and 130–238 environment steps, respectively, with 30 held-out states per bucket. One world-model step corresponds to five environment steps. The world model and planner remain fixed, including the prediction horizon h , as task duration increases.

Table II compares all variants. Policy only is reported for policies fine-tuned on 5 to 30 demonstrations; PILOT and Flat WM + prior use the 30-demonstration policy as prior. Flat WM achieves 33.3% success on short tasks and 0.0% on both medium and long tasks. Policy initialization improves its short-task success to 46.7%, but medium- and long-task success remain at 0.0%. Under the tested planning budget, the prior therefore improves final-goal action search without substituting for intermediate visual targets on longer tasks.

Direct policy execution improves steadily with demonstrations but degrades with task length at every data level: with 30 demonstrations it reaches 40.0%, 33.3%, and 23.3%

TABLE II
PUSH-T SUCCESS RATE BY TASK LENGTH (30 HELD-OUT INITIAL STATES PER BUCKET).

Method	Short	Medium	Long
Flat WM	33.3%	0.0%	0.0%
Flat WM + prior	46.7%	0.0%	0.0%
Policy only (5 demos)	0.0%	0.0%	0.0%
Policy only (10 demos)	13.3%	3.3%	6.7%
Policy only (20 demos)	26.7%	16.7%	10.0%
Policy only (30 demos)	40.0%	33.3%	23.3%
PILOT w/o prior	43.3%	36.7%	40.0%
PILOT	80.0%	63.3%	53.3%
PILOT (oracle)	90.0%	93.3%	83.3%

on short, medium, and long tasks. PILOT achieves 80.0%, 63.3%, and 53.3%, the highest rates among methods without expert subgoals at every task length, exceeding direct execution of the same 30-demonstration policy by 40.0, 30.0, and 30.0 percentage points. Relative to PILOT w/o prior (43.3%, 36.7%, and 40.0%), the prior adds 36.7, 26.6, and 13.3 points, demonstrating its complementary benefit when intermediate targets are available. Fig. 6 shows a successful execution.

Planning efficiency. Fig. 7(a) sweeps the CEM budget (iterations and samples) at a fixed prediction horizon and subgoal schedule and plots medium-task success against the measured mean time per planning call, for generic initialization and for priors fine-tuned on 10, 20, and 30 demonstrations, the last being the prior used in Table II. Every prior shifts the curve upward, and stronger priors saturate sooner: the 30-demonstration prior exceeds 60% success at 8 s per call, whereas generic initialization reaches 33% at 50 s; even the 10-demonstration prior, whose policy alone succeeds on 3.3% of medium tasks, raises the fitted plateau from 34% to 48%. The policy query itself adds under 0.2 s per call. Panel (b) compares the two initializations on identical planning problems: starting from similar first-iteration costs, policy initialization reaches in four iterations the cost that generic initialization attains in ten, and finishes 24.1% lower under the same sampling budget.

Generated versus oracle subgoals. PILOT (oracle) selects K frames from the corresponding held-out expert trajectory using the same normalized temporal sampling rule as the generated video. Both variants share K , per-subgoal execution budget R , prediction horizon, policy prior, and CEM settings. Oracle-guided success is 90.0%, 93.3%, and 83.3%; the reported rates exceed PILOT by 10.0, 30.0, and 30.0 percentage points, respectively. These gains show room for improving the generated targets under the same controller. A possible source of the gap is that a generated video may not depict successful task completion; visual fidelity, target reachability, and temporal alignment can also affect execution.

H. LIBERO-10: Long-Horizon Manipulation

We evaluate all six variants on 5 held-out initial states per LIBERO-10 task. Table III reports success rates aggregated over the ten tasks, and Fig. 8 shows a successful execution.

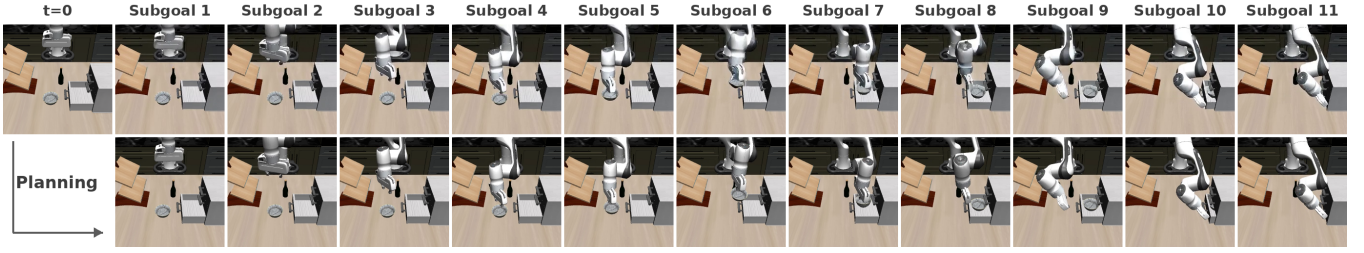


Fig. 8. PILOT on the LIBERO-10 task *put the black bowl in the bottom drawer of the cabinet and close it*. Top: initial observation and video-predicted visual subgoals. Bottom: the simulator observation when each subgoal expires.

TABLE III

LIBERO-10 SUCCESS RATE (5 HELD-OUT INITIAL STATES PER TASK).

Method	Success rate
Flat WM	0.0%
Flat WM + prior	0.0%
Policy only	38.0%
PILOT w/o prior	22.0%
PILOT	60.0%
PILOT (oracle)	88.0%

Flat world-model planning achieves 0.0% success with either initialization. Generated subgoals with generic CEM initialization reach 22.0%, below the 38.0% of direct policy execution. Combining subgoals and the policy prior, PILOT reaches 60.0%, a 22 percentage-point gain over Policy only. Expert subgoals raise success to 88.0%, indicating further gains from improving temporal guidance under the same controller.

IV. CONCLUSIONS

We presented **PILOT**, which executes long manipulation tasks through video-guided, policy-initialized local world-model planning. With Wan and π_0 fine-tuned on the same 40 real-robot demonstrations, PILOT achieves 68.8% success, compared with 55.0% without the prior and 13.8% for direct policy execution. These results show that visual predictions learned from limited demonstrations can guide effective online control when the policy remains unreliable on its own.

PILOT also exceeds direct execution of its own policy prior by 30 percentage points on long Push-T tasks and by 22 points on LIBERO-10. The CEM analysis supports more efficient refinement with policy initialization, with stronger priors reaching a given success rate in less planning time, while expert-subgoal comparisons identify room for improving temporal guidance under the same controller.

V. LIMITATIONS AND FUTURE WORK

PILOT can fail when generated subgoals are unreachable, world-model errors misrank actions, or a narrow policy prior limits exploration. Its fixed subgoal schedule cannot accommodate delayed interactions, which can hinder recovery after a failed grasp. Video generation and online optimization also add latency relative to policy execution. Our real-world evaluation is limited to four tabletop tasks on a UR5e. Future work will investigate adaptive subgoal timing and search distributions, cheaper world-model rollouts, and generalization to more diverse tasks and embodiments.

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